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Command Prompt - mysql x Windows PowerShell x Windows PowerShell x + v

KD4_98836_Gauri@localhost> USE classwork;
Database changed
KD4_98836_Gauri@localhost> DROP TABLE IF EXISTS allocations;
Query OK, 0 rows affected, 1 warning (0.03 sec)

KD4_98836_Gauri@localhost> DROP TABLE IF EXISTS preferences;
Query OK, 0 rows affected, 1 warning (0.00 sec)

KD4_98836_Gauri@localhost> DROP TABLE IF EXISTS student_ranks;
Query OK, 0 rows affected, 1 warning (0.02 sec)

KD4_98836_Gauri@localhost> DROP TABLE IF EXISTS center_courses;
Query OK, 0 rows affected, 1 warning (0.00 sec)

KD4_98836_Gauri@localhost> DROP TABLE IF EXISTS courses;
Query OK, 0 rows affected, 1 warning (0.02 sec)

KD4_98836_Gauri@localhost> DROP TABLE IF EXISTS centers;
Query OK, 0 rows affected, 1 warning (0.02 sec)

KD4_98836_Gauri@localhost> DROP TABLE IF EXISTS students;
Query OK, 0 rows affected, 1 warning (0.00 sec)

KD4_98836_Gauri@localhost> -- 1. TABLE CREATION WITH CASCADE CONSTRAINTS
Query OK, 0 rows affected (0.00 sec)

KD4_98836_Gauri@localhost> CREATE TABLE students (
-> student_id INT PRIMARY KEY AUTO_INCREMENT,
-> name VARCHAR(100) NOT NULL,
-> email VARCHAR(100) UNIQUE NOT NULL,
-> mobile VARCHAR(15) NOT NULL
-> );
Query OK, 0 rows affected (0.10 sec)

KD4_98836_Gauri@localhost> CREATE TABLE centers (
-> center_id INT PRIMARY KEY AUTO_INCREMENT,
-> center_code VARCHAR(10) UNIQUE NOT NULL,
-> center_name VARCHAR(100) NOT NULL,
-> city VARCHAR(50) NOT NULL
-> );
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Command Prompt - mysql x Windows PowerShell x Windows PowerShell x + v

KD4_98836_Gauri@localhost> -- Center-Course Capacities
Query OK, 0 rows affected (0.00 sec)

KD4_98836_Gauri@localhost> CREATE TABLE center_courses (
-> center_course_id INT PRIMARY KEY AUTO_INCREMENT,
-> center_id INT NOT NULL,
-> course_id INT NOT NULL,
-> capacity INT NOT NULL,
-> filled_seats INT DEFAULT 0
-> FOREIGN KEY (center_id) REFERENCES centers(center_id) ON DELETE CASCADE ON UPDATE CASCADE,
-> FOREIGN KEY (course_id) REFERENCES courses(course_id) ON DELETE CASCADE ON UPDATE CASCADE,
-> CONSTRAINT unique_center_course UNIQUE (center_id, course_id)
-> );
Query OK, 0 rows affected (0.05 sec)

KD4_98836_Gauri@localhost> -- Student C-CAT Ranks
Query OK, 0 rows affected (0.00 sec)

KD4_98836_Gauri@localhost> CREATE TABLE student_ranks (
-> student_id INT PRIMARY KEY,
-> ccat_rank INT NOT NULL UNIQUE,
-> FOREIGN KEY (student_id) REFERENCES students(student_id) ON DELETE CASCADE ON UPDATE CASCADE
-> );
Query OK, 0 rows affected (0.04 sec)

KD4_98836_Gauri@localhost> -- Student Choice Preferences
Query OK, 0 rows affected (0.00 sec)

KD4_98836_Gauri@localhost> CREATE TABLE preferences (
-> preference_id INT PRIMARY KEY AUTO_INCREMENT,
-> student_id INT NOT NULL,
-> center_course_id INT NOT NULL,
-> preference_number INT NOT NULL,
-> FOREIGN KEY (student_id) REFERENCES students(student_id) ON DELETE CASCADE ON UPDATE CASCADE,
-> FOREIGN KEY (center_course_id) REFERENCES center_courses(center_course_id) ON DELETE CASCADE ON UPDATE CASCADE,
-> CONSTRAINT unique_student_pref UNIQUE (student_id, preference_number)
-> );
Query OK, 0 rows affected (0.04 sec)

KD4_98836_Gauri@localhost> -- Final Output Allocations
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Command Prompt - mysql x Windows PowerShell x Windows PowerShell x + v
-> ('VITA', 'VITA Juhu', 'Mumbai');
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0

KD4_98836_Gauri@localhost> INSERT INTO courses (course_code, course_name) VALUES
-> ('DAC', 'PG Diploma in Advanced Computing'),
-> ('DMC', 'PG Diploma in Mobile Computing');
Query OK, 2 rows affected (0.01 sec)
Records: 2 Duplicates: 0 Warnings: 0

KD4_98836_Gauri@localhost> -- Map Options and Mock Capacities
Query OK, 0 rows affected (0.00 sec)

KD4_98836_Gauri@localhost> INSERT INTO center_courses (center_id, course_id, capacity) VALUES
-> (2, 1, 2), -- ID 1: ACTS - DAC (Capacity: 2)
-> (1, 1, 1), -- ID 2: IACSD - DAC (Capacity: 1)
-> (3, 1, 2), -- ID 3: VITA - DAC (Capacity: 2)
-> (2, 2, 1); -- ID 4: ACTS - DMC (Capacity: 1)
Query OK, 4 rows affected (0.02 sec)
Records: 4 Duplicates: 0 Warnings: 0

KD4_98836_Gauri@localhost> -- Assign Merit Ranks (Lower value means superior rank performance)
Query OK, 0 rows affected (0.00 sec)

KD4_98836_Gauri@localhost> INSERT INTO student_ranks (student_id, ccat_rank) VALUES
-> (1, 10), (2, 25), (3, 40), (4, 55), (5, 120);
Query OK, 5 rows affected (0.03 sec)
Records: 5 Duplicates: 0 Warnings: 0

KD4_98836_Gauri@localhost> -- Map Preferences Choice Options
Query OK, 0 rows affected (0.00 sec)

KD4_98836_Gauri@localhost> INSERT INTO preferences (student_id, center_course_id, preference_number) VALUES
-> (1, 1, 1), (1, 2, 2), -- Amit: [1] ACTS-DAC, [2] IACSD-DAC
-> (2, 1, 1), (2, 2, 2), -- Priya: [1] ACTS-DAC, [2] IACSD-DAC
-> (3, 1, 1), (3, 2, 2),
-> (4, 2, 1), (4, 4, 2),
-> (5, 1, 1);
Query OK, 9 rows affected (0.01 sec)
Records: 9 Duplicates: 0 Warnings: 0
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Command Prompt - mysql x Windows PowerShell x Windows PowerShell x + v
KD4_98836_Gauri@localhost> DELIMITER //
KD4_98836_Gauri@localhost> CREATE PROCEDURE ProcessRound1Allocation()
-> BEGIN
-> DECLARE done INT DEFAULT FALSE;
-> DECLARE current_student_id INT;
-> DECLARE current_pref_cc_id INT;
->
-> DECLARE student_cursor CURSOR FOR
-> SELECT r.student_id FROM student_ranks r ORDER BY r.ccat_rank ASC;
-> DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = TRUE;
->
-> -- Clean old run tracks safely
-> SET FOREIGN_KEY_CHECKS = 0;
-> TRUNCATE TABLE allocations;
-> SET FOREIGN_KEY_CHECKS = 1;
->
-> UPDATE center_courses SET filled_seats = 0;
->
-> OPEN student_cursor;
-> student_loop: LOOP
-> FETCH student_cursor INTO current_student_id;
-> IF done THEN LEAVE student_loop; END IF;
->
-> BEGIN
-> DECLARE pref_done INT DEFAULT FALSE;
-> DECLARE allocated INT DEFAULT FALSE;
-> DECLARE pref_cursor CURSOR FOR
-> SELECT p.center_course_id FROM preferences p
-> WHERE p.student_id = current_student_id ORDER BY p.preference_number ASC;
-> DECLARE CONTINUE HANDLER FOR NOT FOUND SET pref_done = TRUE;
->
-> OPEN pref_cursor;
-> pref_loop: LOOP
-> FETCH pref_cursor INTO current_pref_cc_id;
-> IF pref_done OR allocated THEN LEAVE pref_loop; END IF;
->
-> -- Check dynamic seat availability counter
-> IF (SELECT (capacity - filled_seats) FROM center_courses WHERE center_course_id = current_pref_cc_id) > 0 THEN
-> INSERT INTO allocations (student_id, center_course_id, allocated_round, allocation_status)
-> VALUES (current_student_id, current_pref_cc_id, 1, 'ALLOCATED');
```

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Command Prompt - mysql x Windows PowerShell x Windows PowerShell x + v
-> SET allocated = TRUE;
-> END IF;
-> END LOOP pref_loop;
-> CLOSE pref_cursor;
-> END;
-> END LOOP student_loop;
-> CLOSE student_cursor;
->
-- FIXED COMPILER MAP: Linked explicitly on your custom primary keys
-> SELECT a.allocation_id, s.name, r.ccat_rank, cen.center_code, c.course_code, a.allocated_round, a.allocation_status
-> FROM allocations a
-> JOIN students s ON a.student_id = s.student_id
-> JOIN student_ranks r ON a.student_id = r.student_id
-> JOIN center_courses cc ON a.center_course_id = cc.center_course_id
-> JOIN centers cen ON cc.center_id = cen.center_id
-> JOIN courses c ON cc.course_id = c.course_id
-> ORDER BY r.ccat_rank ASC;
-> END //
Query OK, 0 rows affected (0.03 sec)

KDH_98836_Gauri@localhost>
KDH_98836_Gauri@localhost> DELIMITER ;
KDH_98836_Gauri@localhost> CALL ProcessRound1Allocation();
+-----+-----+-----+-----+-----+-----+-----+
| allocation_id | name      | ccat_rank | center_code | course_code | allocated_round | allocation_status |
+-----+-----+-----+-----+-----+-----+-----+
| 1             | Amit Sharma | 10        | ACTS        | DAC         | 1              | ALLOCATED         |
| 2             | Priya Patel  | 25        | ACTS        | DAC         | 1              | ALLOCATED         |
| 3             | Rahul Verma  | 40        | IACSD       | DAC         | 1              | ALLOCATED         |
| 4             | Sneha Reddy  | 55        | ACTS        | DMC         | 1              | ALLOCATED         |
+-----+-----+-----+-----+-----+-----+-----+
4 rows in set (0.08 sec)

Query OK, 0 rows affected (0.09 sec)

KDH_98836_Gauri@localhost> DELIMITER //
KDH_98836_Gauri@localhost>
KDH_98836_Gauri@localhost> CREATE PROCEDURE ProcessRound2Allocation()
-> BEGIN
-> DECLARE done INT DEFAULT FALSE;
-> DECLARE current_student_id INT;
```

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Command Prompt - mysql x Windows PowerShell x Windows PowerShell x + v
KDH_98836_Gauri@localhost> CREATE PROCEDURE ProcessRound2Allocation()
-> BEGIN
-> DECLARE done INT DEFAULT FALSE;
-> DECLARE current_student_id INT;
-> DECLARE current_rank INT;
-> DECLARE current_allocated_cc_id INT;
-> DECLARE current_allocated_pref_num INT;
-> DECLARE better_pref_cc_id INT;
->
-> DECLARE student_cursor CURSOR FOR
-> SELECT r.student_id, r.ccat_rank FROM student_ranks r ORDER BY r.ccat_rank ASC;
-> DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = TRUE;
->
-> OPEN student_cursor;
-> student_loop: LOOP
-> FETCH student_cursor INTO current_student_id, current_rank;
-> IF done THEN LEAVE student_loop; END IF;
->
-> SET current_allocated_cc_id = NULL;
->
-> SELECT center_course_id INTO current_allocated_cc_id
-> FROM allocations WHERE student_id = current_student_id LIMIT 1;
->
-> IF current_allocated_cc_id IS NOT NULL THEN
-> -- Upgradation Logic
-> SELECT preference_number INTO current_allocated_pref_num FROM preferences
-> WHERE student_id = current_student_id AND center_course_id = current_allocated_cc_id;
->
-> BEGIN
-> DECLARE upgrade_done INT DEFAULT FALSE;
-> DECLARE upgraded INT DEFAULT FALSE;
-> DECLARE upgrade_cursor CURSOR FOR
-> SELECT p.center_course_id FROM preferences p
-> WHERE p.student_id = current_student_id AND p.preference_number < current_allocated_pref_num
-> ORDER BY p.preference_number ASC;
-> DECLARE CONTINUE HANDLER FOR NOT FOUND SET upgrade_done = TRUE;
->
-> OPEN upgrade_cursor;
-> upgrade_loop: LOOP
-> FETCH upgrade_cursor INTO better_pref_cc_id;
-> IF upgrade_done OR upgraded THEN LEAVE upgrade_loop; END IF;
```

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Command Prompt - mysql x Windows PowerShell x Windows PowerShell x + v
->
-> BEGIN
-> DECLARE upgrade_done INT DEFAULT FALSE;
-> DECLARE upgraded INT DEFAULT FALSE;
-> DECLARE upgrade_cursor CURSOR FOR
-> SELECT p.center_course_id FROM preferences p
-> WHERE p.student_id = current_student_id AND p.preference_number < current_allocated_pref_num
-> ORDER BY p.preference_number ASC;
-> DECLARE CONTINUE HANDLER FOR NOT FOUND SET upgrade_done = TRUE;
->
-> OPEN upgrade_cursor;
-> upgrade_loop: LOOP
-> FETCH upgrade_cursor INTO better_pref_cc_id;
-> IF upgrade_done OR upgraded THEN LEAVE upgrade_loop; END IF;
->
-> IF (SELECT (capacity - filled_seats) FROM center_courses WHERE center_course_id = better_pref_cc_id) > 0 THEN
-> UPDATE center_courses SET filled_seats = filled_seats - 1 WHERE center_course_id = current_allocated_cc_id;
-> UPDATE allocations SET center_course_id = better_pref_cc_id, allocated_round = 2, allocation_status = 'UPGRADED' WHERE studen
t_id = current_student_id;
-> UPDATE center_courses SET filled_seats = filled_seats + 1 WHERE center_course_id = better_pref_cc_id;
-> SET upgraded = TRUE;
-> END IF;
-> END LOOP upgrade_loop;
-> CLOSE upgrade_cursor;
-> END;
-> ELSE
-> -- Allocation configuration for previously unallocated students
-> BEGIN
-> DECLARE fresh_done INT DEFAULT FALSE;
-> DECLARE fresh_allocated INT DEFAULT FALSE;
-> DECLARE fresh_cursor CURSOR FOR
-> SELECT p.center_course_id FROM preferences p WHERE p.student_id = current_student_id ORDER BY p.preference_number ASC;
-> DECLARE CONTINUE HANDLER FOR NOT FOUND SET fresh_done = TRUE;
->
-> OPEN fresh_cursor;
-> fresh_loop: LOOP
-> FETCH fresh_cursor INTO better_pref_cc_id;
-> IF fresh_done OR fresh_allocated THEN LEAVE fresh_loop; END IF;
->
-> IF (SELECT (capacity - filled_seats) FROM center_courses WHERE center_course_id = better_pref_cc_id) > 0 THEN
-> INSERT INTO allocations (student_id, center_course_id, allocated_round, allocation_status)
```

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Command Prompt - mysql x Windows PowerShell x Windows PowerShell x + v
-> CLOSE student_cursor;
->
-> -- FIXED COMPILER MAP: Swapped 's.id' out for your explicit 'student_id' naming structure
-> SELECT a.allocation_id, s.name, r.ccat_rank, cen.center_code, c.course_code, a.allocated_round, a.allocation_status
-> FROM allocations a
-> JOIN students s ON a.student_id = s.student_id
-> JOIN student_ranks r ON s.student_id = r.student_id
-> JOIN center_courses cc ON a.center_course_id = cc.center_course_id
-> JOIN centers cen ON cc.center_id = cen.center_id
-> JOIN courses c ON cc.course_id = c.course_id
-> ORDER BY r.ccat_rank ASC;
-> END //
Query OK, 0 rows affected (0.01 sec)

KD4_98836_Gauri@localhost>
KD4_98836_Gauri@localhost> DELIMITER ;
KD4_98836_Gauri@localhost> CALL ProcessRound2Allocation();

+-----+-----+-----+-----+-----+-----+-----+
| allocation_id | name      | ccat_rank | center_code | course_code | allocated_round | allocation_status |
+-----+-----+-----+-----+-----+-----+-----+
| 1 | Amit Sharma | 10 | ACTS | DAC | 1 | ALLOCATED |
| 2 | Priya Patel | 25 | ACTS | DAC | 1 | ALLOCATED |
| 3 | Rahul Verma | 40 | IACSD | DAC | 1 | ALLOCATED |
| 4 | Sneha Reddy | 55 | ACTS | DMC | 1 | ALLOCATED |
+-----+-----+-----+-----+-----+-----+-----+

4 rows in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)

KD4_98836_Gauri@localhost>
```